

# GRANDSTAFF, CHRISTINE & NORMAN

1065 22nd St, Sarasota,  
Florida, 34234

## GENERAL NOTES

### 1. DESIGN CRITERIA

FLORIDA BUILDING CODE (FBC), 8TH EDITION (2023/2024).  
ASCE 7-22: MINIMUM DESIGN LOADS AND ASSOCIATED CRITERIA FOR BUILDINGS AND OTHER STRUCTURES.  
ACI 318-19: BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE.  
ASTM STANDARDS AS SPECIFIED HEREIN.  
ALUMINUM DESIGN MANUAL 2020 EDITION (AA ADM-2020).  
AA ASM 35-30: ALUMINUM STANDARDS AND DATA.  
NATIONAL ELECTRICAL CODE (NEC), 2023 EDITION.

### 2. DESIGN LOADS

WIND SPEED (FBC CHAPTER 16): 140 MPH  
ULTIMATE WIND SPEED (VULT) = 138 MPH (PER ASCE 7-22, SECTION 26.5.1).  
ALLOWABLE STRESS DESIGN WIND SPEED (VASD) = 107 MPH (VASD = VULT × √0.6).  
RISK CATEGORY: I (LOW HAZARD TO HUMAN LIFE).  
EXPOSURE CATEGORY: B (URBAN OR SUBURBAN TERRAIN WITH OBSTRUCTIONS).  
DESIGN WIND PRESSURES INTERPOLATED FROM ASCE 7-22 AND REFERENCE TABLES FOR VULT = 140 MPH, EXPOSURE CATEGORY B.

WIND LOADS (MWFRS), BASED ON 140 MPH WIND SPEED AND EXPOSURE B:

- HORIZONTAL PRESSURE (WINDWARD): 26 PSF.
- HORIZONTAL PRESSURE (LEEWARD): 21 PSF.
- VERTICAL PRESSURE ON SCREEN SURFACE: 8 PSF.
- VERTICAL PRESSURE ON SOLID SURFACES: 27 PSF.

LOAD REDUCTION FACTORS:

- FOR 20x20x.011" MESH SCREEN: 0.88 (MANUFACTURER DATA, ASCE 7-22 SECTION 30.5).
- FOR ALLOWABLE STRESS DESIGN PRESSURES: 0.6 (ASCE 7-22 SECTION 2.4).

### 3. CONTRACTOR RESPONSIBILITIES

- ALL WORK SHALL BE PERFORMED BY A LICENSED CONTRACTOR IN ACCORDANCE WITH APPLICABLE CODES, REGULATIONS, AND LOCAL ORDINANCES.
- CONTRACTOR MUST FIELD-VERIFY ALL DIMENSIONS PRIOR TO FABRICATION OR CONSTRUCTION. WRITTEN DIMENSIONS SHALL GOVERN OVER SCALED DRAWINGS.
- ANY DISCREPANCIES, ERRORS, OR OMISSIONS SHALL BE REPORTED TO THE ENGINEER IN WRITING WITHIN 10 DAYS OF PLAN RECEIPT AND PRIOR TO CONSTRUCTION.
- CONTRACTOR IS RESPONSIBLE FOR VERIFYING EXISTING SITE CONDITIONS, INCLUDING STRUCTURE AND FOUNDATION INTEGRITY, PRIOR TO COMMENCEMENT.
- CONTRACTOR SHALL PROVIDE ALL REQUIRED TEMPORARY SHORING, BRACING, AND SAFETY MEASURES DURING CONSTRUCTION.
- CONTRACTOR SHALL PROTECT EXISTING STRUCTURES, UTILITIES, AND LANDSCAPING OUTSIDE THE SCOPE OF THIS PROJECT.
- INSTALLATION PRACTICES MUST COMPLY WITH MANUFACTURER RECOMMENDATIONS, INCLUDING FLASHING AND WATERPROOFING DETAILS.
- CONTRACTOR SHALL COORDINATE SITE ACCESS FOR INSPECTIONS AND MAINTAIN CLEAN AND SAFE WORKING CONDITIONS.
- ALL UNDERGROUND UTILITIES SHALL BE LOCATED AND PROTECTED PRIOR TO EXCAVATION.
- SHOP DRAWINGS AND MATERIAL CERTIFICATIONS SHALL BE PROVIDED UPON REQUEST.
- CONTRACTOR SHALL MAINTAIN RECORDS OF INSPECTIONS AND TESTING.
- FASTENER TORQUE AND EMBEDMENT TESTING RESULTS SHALL BE PROVIDED IF REQUESTED.

### 4. MATERIALS

#### FOUNDATION

- ASSUMED SOIL BEARING CAPACITY: 2,000 PSF (VERIFY IN FIELD).
- NOTIFY ENGINEER IF UNSUITABLE SOILS ARE ENCOUNTERED.
- APPLY TERMITE TREATMENT AS REQUIRED BY FBC.
- PLACE CONCRETE OVER VAPOR BARRIER WHERE APPLICABLE.

#### CONCRETE

- MINIMUM COMPRESSIVE STRENGTH: 3,000 PSI @ 28 DAYS.
- PORTLAND CEMENT: ASTM C150, TYPE I.

- AGGREGATE: 3/4" MAXIMUM, ASTM C33.
- AIR ENTRAINMENT: 4–7%, ASTM C260.
- WATER-REDUCING AGENT: ASTM C494.
- WATER: CLEAN, POTABLE.
- MINIMUM SLAB THICKNESS: 4 INCHES.
- CURE CONCRETE FOR A MINIMUM OF 7 DAYS.

#### STEEL REINFORCEMENT

- REINFORCING STEEL: ASTM A615, GRADE 60.
- MINIMUM COVER FOR FOOTINGS: 2 INCHES.
- LAP SPLICES: 40 TIMES BAR DIAMETER.
- WELDED WIRE MESH PER ASTM A185.
- FIBERMESH MAY BE SUBSTITUTED IF ASTM AND ACI REQUIREMENTS ARE MET.

#### ALUMINUM

- STRUCTURAL ALUMINUM ALLOY: 6005-T5 MINIMUM.
- DESIGN PER AA ADM-2020 SPECIFICATIONS.
- MINIMUM WALL THICKNESS FOR EXTRUDED MEMBERS: 0.040 INCHES.
- SELF-MATING BEAMS TO BE STITCHED WITH #12 SMS SCREWS:
  - 6" O.C. AT ENDS.
  - 18" O.C. ALONG SPANS.
- KICK PLATES (FORMED PANELS) MINIMUM THICKNESS: 0.024 INCHES.
- 18X14X0.013" MESH SCREEN IS STANDARD UNLESS OTHERWISE APPROVED.
- ROOF BRACING: 2" X 2" X 0.050" WALL THICKNESS MINIMUM.
- ALL ALUMINUM IN CONTACT WITH STEEL OR PRESSURE-TREATED LUMBER SHALL HAVE DIELECTRIC ISOLATION.
- ALUMINUM ROOF SYSTEMS SHALL BE SELECTED PER MANUFACTURER SPAN TABLES CERTIFIED BY A LICENSED FLORIDA ENGINEER.
- METAL STRUCTURES WITHIN 5 FEET OF POOLS SHALL BE BONDED PER NEC 680.26.

#### FASTENERS

- ALL FASTENERS SHALL COMPLY WITH ASTM A153 FOR ZINC COATING.
- CONCRETE SCREWS: TAPCON, SIMPSON, HILTI, REDHEAD, OR APPROVED EQUAL.
- LAG BOLTS SHALL HAVE MINIMUM EMBEDMENT OF 8 TIMES BOLT DIAMETER.
- STAINLESS STEEL TYPE 300 (18-8) SHALL BE USED FOR CONNECTIONS TO PRESSURE-TREATED WOOD.
- MINIMUM FASTENER SPACING:
  - SMS: 3/4" CENTER-TO-CENTER, 1/2" EDGE DISTANCE.
  - 1X2 MEMBERS TO MASONRY: 1/4" X 2-3/4" TAPCONS @ 18" O.C.
  - 1X2 MEMBERS TO WOOD: #14 X 2-3/4" WOOD SCREWS @ 18" O.C.
  - 1X2 MEMBERS TO ALUMINUM: #10 X 1-1/2" SMS OR TEK SCREWS @ 18" O.C.
- FOR "OPEN VIEW" OR "PICTURE WINDOW" ASSEMBLIES, REDUCE FASTENER SPACING TO 12" O.C.

### 5. ELECTRICAL

- METAL STRUCTURES WITHIN 5 FEET OF SWIMMING POOLS MUST BE BONDED IN ACCORDANCE WITH NEC 680.26.
- PROVIDE EQUIPOTENTIAL BONDING FOR ALL CONDUCTIVE ELEMENTS WITHIN POOL AREA.
- ALL ELECTRICAL WORK SHALL COMPLY WITH NEC 2023 EDITION AND LOCAL AMENDMENTS.

### 6. GENERAL REQUIREMENTS

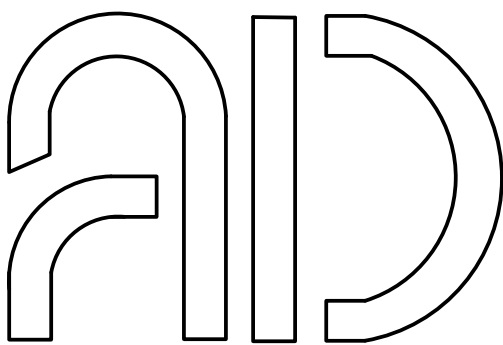
- DIMENSIONS ARE NOMINAL UNLESS OTHERWISE NOTED.
- DIMENSIONAL TOLERANCES: ±1/8" UNLESS SPECIFIED.
- NOTES APPLY TO THE ENTIRE DRAWING SET UNLESS OTHERWISE INDICATED.
- REPORT ANY CONFLICTS IN DRAWINGS TO THE ENGINEER IMMEDIATELY.
- PROVIDE SHOP DRAWINGS AND MATERIAL CERTIFICATIONS IF REQUESTED.
- ENSURE SITE ACCESS FOR REQUIRED INSPECTIONS.
- FINAL CLEANUP IS REQUIRED PRIOR TO PROJECT CLOSEOUT.

| ALUMINUM EXTRUSION DIMENSIONS |                            |
|-------------------------------|----------------------------|
| HOLLOW SECTIONS               |                            |
| 2 x 2                         | 2" x 2" x 0.050"           |
| 2 x 3                         | 2" x 3" x 0.050"           |
| OPEN-BACK SECTIONS            |                            |
| 1 x 2                         | 1" x 2" x 0.044"           |
| SELF-MATING SECTIONS (SMB)    |                            |
| 2 x 4                         | 2" x 4" x 0.044" x 0.100"  |
| 2 x 5CL                       | 2" x 5" x 0.050" x 0.135"  |
| 2 x 6                         | 2" x 6" x 0.050" x 0.120"  |
| 2 x 7                         | 2" x 7" x 0.057" x 0.120"  |
| 2 x 8                         | 2" x 8" x 0.072" x 0.224"  |
| 2 x 9                         | 2" x 9" x0.072" x 0.224"   |
| 2 x 10                        | 2" x 10" x 0.092" x 0.374" |
| REC-TUBE - 6005-T5 MINI.      |                            |
| 2 x 8                         | 2" x 8" x 0.125" x 0.125"  |

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1

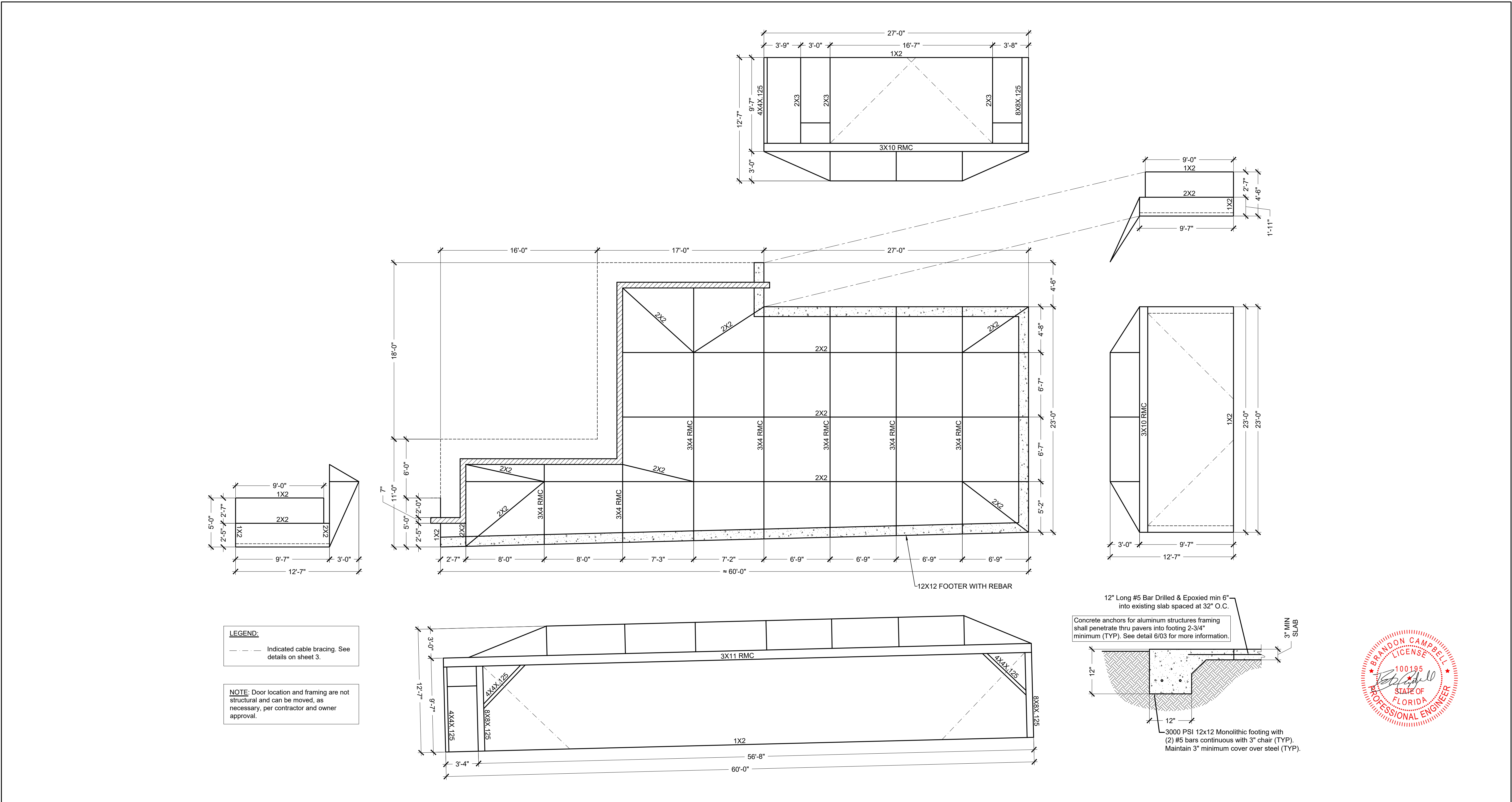
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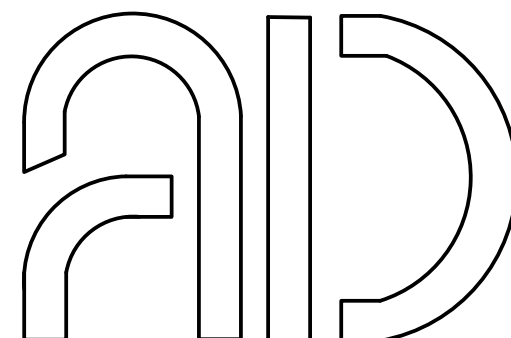
NOTES

#### PROJECT NAME:

GRANDSTAFF, CHRISTINE & NORMAN





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|-----|----------------|-------|----------|-------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------|
|     |                |       |          |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                    | <b>DRAWING TITLE:</b><br>STRUCTURAL DESIGN             |                    |
|     |                |       |          |       |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                    | <b>PROJECT NAME:</b><br>GRANDSTAFF, CHRISTINE & NORMAN |                    |
| 1   | INITIAL DESIGN | JDGV  | AL       | BC    | 5/29/2026 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                    |                                                        |                    |
| REV | DESCRIPTION    | DRAWN | REVIEWED | APPR. | DATE      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                    |                                                        |                    |



**CABLE TO FRAME CONNECTION**  
SCALE: N.T.S.

**SLOPED PURLIN TO COLUMN CONNECTION**  
SCALE: N.T.S.

**CABLE TO FOUNDATION CONNECTION**  
SCALE: N.T.S.

**ROOF BRACE CONNECTION**  
SCALE: N.T.S.

**COLUMN TO FND CONNECTION FOR 2 x 4 OR LARGER**  
SCALE: N.T.S.

**BEAM TO GUTTER CONNECTION**  
SCALE: N.T.S.

**COLUMN TO BEAM CONNECTION**  
SCALE: N.T.S.

| BEAM SIZE | QTY-SIZE SMS/SIDE |
|-----------|-------------------|
| 2X4 SMB   | 4 - #10           |
| 2X5 SMB   | 4 - #10           |
| 2X6 SMB   | 4 - #10           |
| 2X7 SMB   | 6 - #10           |
| 2X8 SMB   | 6 - #10           |
| 2X9 SMB   | 6 - #10           |
| 2X10 SMB  | 6 - #10           |
| 3X4 RMC   | 4 - #10           |
| 3X5 RMC   | 4 - #10           |
| 3X6 RMC   | 4 - #10           |
| 3X7 RMC   | 6 - #10           |
| 3X8 RMC   | 6 - #10           |
| 3X9 RMC   | 6 - #10           |
| 3X10 RMC  | 6 - #10           |
| 3X11 RMC  | 6 - #10           |
| 4X10 RMC  | 6 - #10           |

**BEAM/POST TO GIRT/PURLIN CONNECTION**  
SCALE: N.T.S.

**GUSSET CONNECTION**  
SCALE: N.T.S.

| BEAM SIZE | QUANTITY OF #12 SMS PER BEAM FACE/SIDE/TOTAL FOR CONNECTION | QUANTITY OF #14 SMS PER BEAM FACE/SIDE/TOTAL FOR CONNECTION |
|-----------|-------------------------------------------------------------|-------------------------------------------------------------|
| 2X4 SMB   | 9/18/36                                                     |                                                             |
| 2X5 SMB   | 9/18/36                                                     |                                                             |
| 2X6 SMB   | 9/18/36                                                     |                                                             |
| 2X7 SMB   | 10/20/40                                                    |                                                             |
| 2X8 SMB   |                                                             | 12/24/48                                                    |
| 2X9 SMB   |                                                             | 12/24/48                                                    |
| 2X9H SMB  |                                                             | 12/24/48                                                    |
| 2X10 SMB  |                                                             | 13/26/52                                                    |
| 3X4 RMC   | 9/18/36                                                     |                                                             |
| 3X5 RMC   | 9/18/36                                                     |                                                             |
| 3X6 RMC   | 9/18/36                                                     |                                                             |
| 3X7 RMC   | 10/20/40                                                    |                                                             |
| 3X8 RMC   |                                                             | 12/24/48                                                    |
| 3X9 RMC   |                                                             | 12/24/48                                                    |
| 3X10 RMC  |                                                             | 13/26/52                                                    |
| 3X11 RMC  |                                                             | 13/26/52                                                    |
| 4X10 RMC  |                                                             | 13/26/52                                                    |

|     |                |       |          |       |           |
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| <b>DRAWING TITLE:</b><br>DETAILS 1                     |                    |
| <b>PROJECT NAME:</b><br>GRANDSTAFF, CHRISTINE & NORMAN |                    |



NOTCH COLUMN TO RECEIVE BEAM. FASTEN EDGE BEAM TO COLUMN WITH (3) 3/8" HEX BOLTS WITH NUT AND LOCK WASHER

EDGE BEAM PER PLAN

PROVIDE 1/8" x 1-1/2" x BEAM HEIGHT ALUMINUM CHANNEL (INTERNAL OR EXTERNAL). FASTEN TO COLUMN WITH (8) #14 x 5/8" SMS. FASTEN EACH SIDE WITH (6) #14 x 5/8" SMS

NOTCH COLUMN TO RECEIVE BEAM. FASTEN EDGE BEAM TO COLUMN WITH (3) 3/8" HEX BOLTS WITH NUT AND LOCK WASHER

PROVIDE 2" x 2" x 1/8" ALUMINUM ANGLE, BEAM HEIGHT. FASTEN EACH LEG WITH (8) #14 x 5/8" SMS (TYP)

EDGE BEAM PER PLAN

**BEAM TO COLUMN CONNECTION**  
SCALE: N.T.S.

PURLIN PER PLAN

0.044" RECEIVING CHANNEL; FASTEN CHANNEL TO SIDE OF BEAM WITH (4) #12 x 5/8" SMS (INTERNAL). FASTEN PURLIN TO CHANNEL WITH (2) #12 x 5/8" SMS EACH SIDE.

EDGE BEAM PER PLAN

**PURLIN TO EDGE BEAM DETAIL**  
SCALE: N.T.S.

PROVIDE 1/8" INTERNAL CHANNEL, CUT TO FIT WITH NO SLIP TOLERANCE. FASTEN WITH (3) #14 x 9/16" SMS EACH SIDE. FASTEN TO CONCRETE WITH (4) 3/8" x 4" TAPCON OR EQUAL. FOR WOOD, USE 1/4" x 2-1/2" GALVANIZED LAG SCREWS IN LIEU OF MASONRY ANCHORS.

**POST BASE CONNECTION**  
SCALE: N.T.S.

30" MIN

BEAM PER PLAN

BRACE PER PLAN

FASTEN BRACE TO COLUMN AND BEAM WITH 0.044" RECEIVING CHANNEL. FASTEN CHANNEL TO BEAM AND COLUMN WITH (3) #12 x 5/8" SMS (INTERNAL). FASTEN CHANNEL TO BRACE WITH #12 x 5/8" SMS (3) PER SIDE

COLUMN PER PLAN

**K-BRACE DETAIL**  
SCALE: N.T.S.

3/16" 5052-H32 GUSSET, CUT TO FIT INSIDE BEAM WITH NO SLIP TOLERANCE

BEAM PER PLAN

EXAMPLE SCREW PATTERN SHOWN: SCREW QUANTITY PER TABLE BELOW

MAINTAIN FASTENER EDGE DISTANCE OF 5/8" AND MINIMUM 3/8" SEPARATION BETWEEN FASTENERS. INSTALL FASTENERS AROUND THE PERIMETER OF THE CONNECTION FIRST, THEN COMPLETE THE REMAINDER IN THE FIELD.

| FASTENER QUANTITY TABLE |               |                         |
|-------------------------|---------------|-------------------------|
| BEAM SIZE               | FASTENER SIZE | QTY PER FACE/SIDE/TOTAL |
| 2X4 SMB                 | #12           | 6 / 12 / 24             |
| 2X5 SMB                 | #12           | 6 / 12 / 24             |
| 2X6 SMB                 | #12           | 10 / 20 / 40            |
| 2X7 SMB                 | #12           | 12 / 24 / 48            |
| 2X8 SMB                 | #14           | 12 / 24 / 48            |
| 2X9 SMB                 | #14           | 18 / 36 / 72            |
| 2X10 SMB                | #14           | 18 / 36 / 72            |
| 3X4 RMC                 | #12           | 6 / 12 / 24             |
| 3X5 RMC                 | #12           | 6 / 12 / 24             |
| 3X6 RMC                 | #12           | 10 / 20 / 40            |
| 3X7 RMC                 | #12           | 12 / 24 / 48            |
| 3X8 RMC                 | #14           | 12 / 24 / 48            |
| 3X9 RMC                 | #14           | 18 / 36 / 72            |
| 3X10 RMC                | #14           | 18 / 36 / 72            |
| 3X11 RMC                | #14           | 18 / 36 / 72            |
| 4X10 RMC                | #14           | 18 / 36 / 72            |

1. BEAM SPLICE LENGTH TO BE 2 x BEAM HEIGHT.  
2. LOCATE BEAM SPLICE AT 1/3 POINTS, STAGGERED AT EACH END OF BEAM.  
3. PROVIDE MINIMUM 1" CAMBER IN BEAM TO REDUCE SELF-WEIGHT DEFLECTION.

**BEAM SPLICE DETAIL**  
SCALE: N.T.S.

1/8" x 2" x 2" ANGLE EACH SIDE OF BEAM (LENGTH = BEAM NOTCH SIZE). FASTEN WITH #12 x 5/8" SMS AT 1/2" O.C. FASTEN TO GUTTER FACE WITH #12 x 5/8" SMS AT 3/4" O.C.

STRUCTURE GUTTER

FASCIA

BEAM PER PLAN

CLOSE TOP OF BEAM WITH RECEIVER CHANNEL; FASTEN WITH #12 x 5/8" SMS, MINIMUM (3) EACH SIDE.

2" x 2" x 1/8" ANGLE EACH SIDE OF BEAM. FASTEN WITH #12 x 5/8" SMS AT 3/4" O.C. EACH SIDE. FASTEN TO MASONRY HOST WITH (2) 1/4" TAPCONS, PROVIDING MINIMUM 2-1/4" EMBEDMENT INTO HOST

1. FASCIA TO BE FASTENED TO RAFTER TAIL USING (1) 1/4" Ø x 4" LAG SCREW.  
2. A 7" GUTTER MAY BE USED, PROVIDED STRAP LENGTH IS INCREASED BY 2".

**BEAM UNDER OVERHANG DETAIL**  
SCALE: N.T.S.

1/8" THICK INTERNAL RECEIVING CHANNEL; FASTEN TO BEAM WITH (8) #14 SMS THROUGH CHANNEL INTO BEAM

BEAM PER PLAN

CAGE BEAM

SIDE VIEW

CAGE BEAM

BEAM PER PLAN

(INTERNAL). FASTEN EACH SIDE OF BEAM FACE TO CHANNEL WITH (8) #14 SMS. MAINTAIN 3/8" SEPARATION BETWEEN FASTENERS AND FROM EDGES OF MEMBERS

**BEAM TO EDGE BEAM DETAIL**  
SCALE: N.T.S.

|     |                |       |          |       |           |
|-----|----------------|-------|----------|-------|-----------|
|     |                |       |          |       |           |
|     |                |       |          |       |           |
| 1   | INITIAL DESIGN | JDGV  | AL       | BC    | 5/29/2026 |
| REV | DESCRIPTION    | DRAWN | REVIEWED | APPR. | DATE      |

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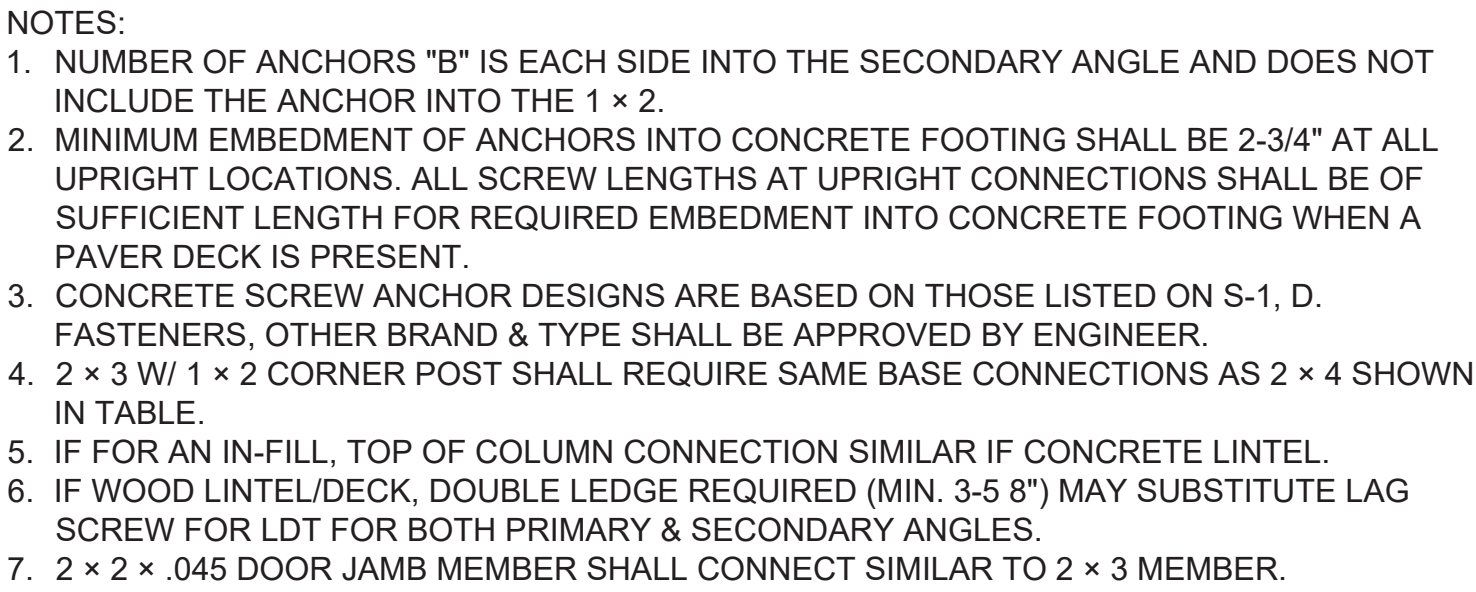
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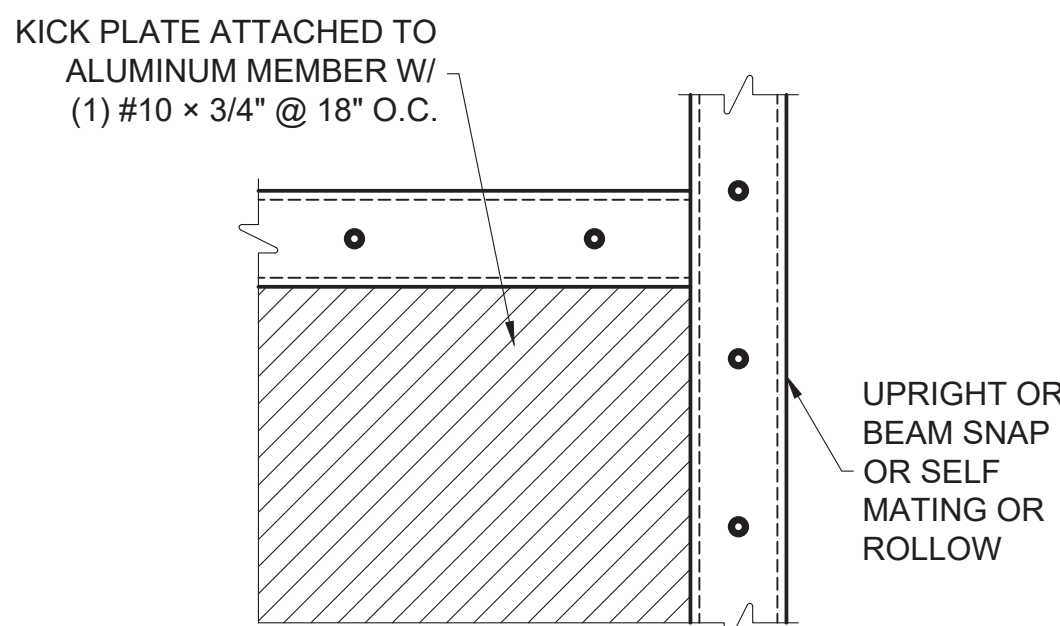
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| <b>DATE:</b><br>5/29/2026                              | <b>SHEET:</b><br>4 |
| <b>DRAWING TITLE:</b><br>DETAILS CONT.                 |                    |
| <b>PROJECT NAME:</b><br>GRANDSTAFF, CHRISTINE & NORMAN |                    |





| COLUMN<br>SIZE | 1/4" AND 3/8" Ø<br>CONCRETE<br>SCREW ANCHOR |                    |
|----------------|---------------------------------------------|--------------------|
|                | "B"                                         | MINIMUM<br>SPACING |
| 2X3 - 1/4"     | 0                                           | 0"                 |
| 2X4 - 1/4"     | 1                                           | 3"                 |
| 2X5 - 1/4"     | 1                                           | 3"                 |
| 2X6 - 3/8"     | 1                                           | 4"                 |
| 2X7 - 3/8"     | 1                                           | 5"                 |
| 2X8 - 3/8"     | 2                                           | 3"                 |
| 2X9 - 3/8"     | 2                                           | 4"                 |
| 2X10 - 3/8"    | 2                                           | 4.5"               |
| 3X4 - 1/4"     | 1                                           | 3"                 |
| 3X5 - 1/4"     | 1                                           | 3"                 |
| 3X6 - 3/8"     | 1                                           | 4"                 |
| 3X7 - 3/8"     | 1                                           | 5"                 |
| 3X8 - 3/8"     | 2                                           | 3"                 |
| 3X9 - 3/8"     | 2                                           | 4"                 |
| 3X10 - 3/8"    | 2                                           | 4.5"               |
| 3X11 - 3/8"    | 2                                           | 4.5"               |
| 4X10 - 3/8"    | 2                                           | 4.5"               |

FOR RECEIVING CHANNEL: TOTAL (8) SCREWS  
(2) #10 x 3/4" SMS (TYP) INTO GIRT/PURLIN &  
(4) #10 x 3/4" SMS (TYP) INTO UPRIGHT/BREAM  
FOR ANGLE: TOTAL (8) SCREWS  
(2) #10 x 3/4" SMS (TYP) @ EA. LEG



**NOTES:**  
KICK PLATE TO BE APPLIED TO UPRIGHT  
CHAIIRRAIL OR GIRT IF APPLICABLE

(4) 3/8" Ø MACHINE BOLTS W/ 3/4" Ø WASHERS 2 × 8 SMB TO 2 × 10 SMB,  
 (3) 3/8" Ø MACHINE BOLTS FOR BEAMS UNDER 2 × 8 SMB, THROUGH CARRIER BEAM AND POST

CARRIER BEAM

ALUMINUM POST

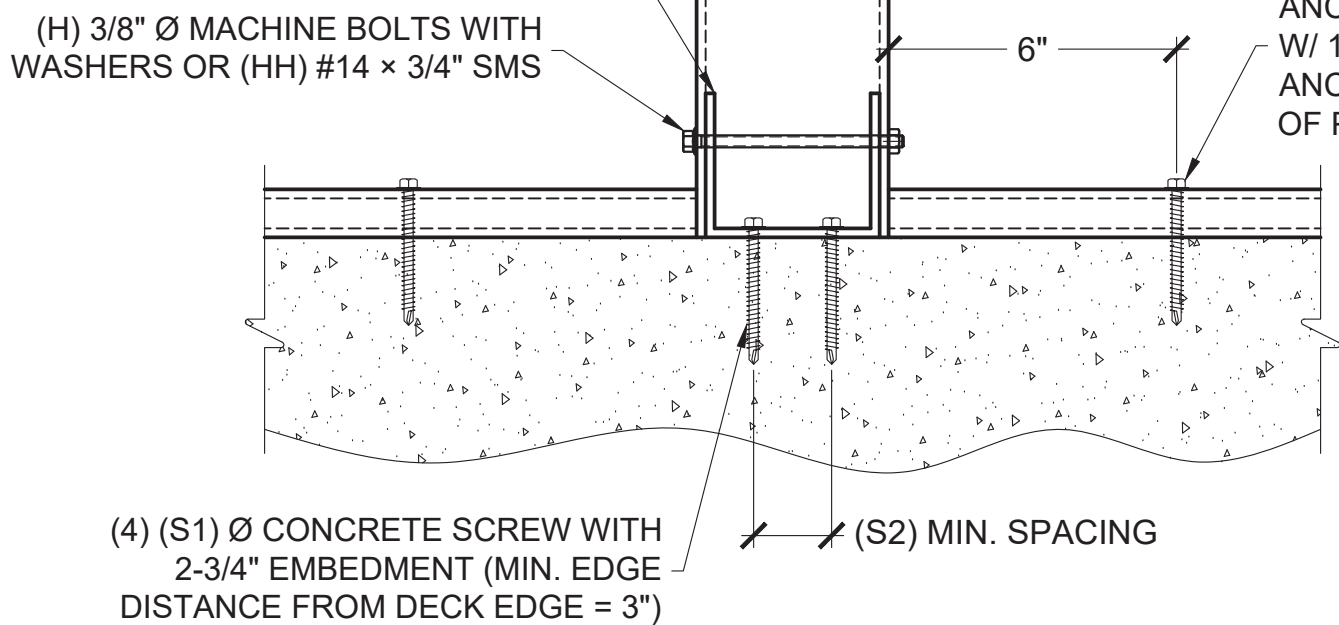
(OPTIONAL) 1" × 2" MEMBER FOR SCREENING  
 USE #10 × 1-1/2" SMS @ 12" O.C. (MAX)  
 CONNECTING 1" × 2" TO CARRIER BEAM & POST

HORIZONTAL BEAM ON ITS SIDE (IF REQUIRED)

**NOTES:**

1. PERPENDICULAR CARRIER BEAM: USE 0.125" RECEIVING CHANNEL WHICH WILL CONNECT TO THE POST W/(4) 3/8" THRU BOLTS USED FOR THE PRIMARY CARRIER BEAM, THEN USE (3) 3/8"Ø MACHINE BOLTS W/ 3/4"Ø WASHERS TO ATTACH THE PERPENDICULAR CARRIER BEAM TO THE RECEIVING CHANNEL.
2. HORIZONTAL BEAM DETAIL WILL BE PROVIDED IF NEEDED.

1" x 2" OPEN BACK  
ANCHORED TO CONCRETE  
W/ 1/4" x 3-1/2" MASONRY  
ANCHOR - 6" MAX EACH SIDE  
OF POST AND 24" O.C. MAX



**NOTES:**

1. CONCRETE SCREW ANCHOR DESIGNS ARE BASED ON TITEN HD (S1) Ø SCREW ANCHORS, OTHER SIZE OR TYPE OF ANCHORS SHALL NOT BE USED.
2. FOR PATIO COVERS AND CARPORTS: DISREGARD THE 1 × 2 OPEN BACK SCREEN MEMBER ON THE FOUNDATION TOP.
3. MINIMUM EMBEDMENT OF ANCHORS INTO CONCRETE FOOTING SHALL BE 2-3/4" AT ALL POST LOCATIONS. ALL SCREW LENGTHS AT POST CONNECTIONS SHALL BE OF SUFFICIENT LENGTH FOR REQUIRED EMBEDMENT INTO CONCRETE FOOTING WHEN A PAVER DECK IS PRESENT.
4. MINIMUM EMBEDMENT OF ANCHORS INTO CONCRETE FOOTING SHALL BE 2-3/4" AT AT ALL UPRIGHT LOCATIONS. ALL SCREW LENGTHS AT UPRIGHT CONNECTIONS SHALL BE OF SUFFICIENT LENGTH FOR REQUIRED EMBEDMENT INTO CONCRETE FOOTING WHEN A PAVER DECK IS PRESENT.
5. DETAIL MAY BE FLIPPED AS NEEDED
6. USE 1/4" × 3" LAG SCREWS IN LIEU OF CONCRETE SCREWS FOR WOOD HEADERS

(8) #14 x 3/4 SMS

2" x 2" x 1/8" EXTRUDED ANGLE ON 45° ANGLE

PERIMETER FRAMING

3/8" Ø ADJUSTABLE EYE BOLT FOR CABLE TENSION

1/8" Ø STAINLESS STEEL CABLE

Technical drawings of the cable support assembly:

- SIDE VIEW:** Shows the side profile of the assembly. Labels include:
  - 1" x 2" x 1/8" ANGLE (4) #12 x 3/4" SMS (TYP TOP AND SIDE)
  - Ø1/8" STAINLESS STEEL CABLE
- TOP VIEW:** Shows the top profile of the assembly. Labels include:
  - 1" Ø (MIN) WASHER BENEATH NUT
  - 3/8" WELDED STAINLESS STEEL EYE BOLT

**NOTES:**  
WHEN USING CORNER CONNECTION OPTION  
#2 CONNECTION CABLE MUST BE WITHIN 6"  
OF CORNER OR PRIMARY MEMBER.

Diagram illustrating the attachment of a 1/8" ALUMINUM PLATE @ 45° ANGLE W/ 1" BREAK @ 90° W/ 5/16" EYE BOLT W/ LOCK NUT & WASHER EA SIDE.

The plate is attached to a wall using (9) #14 x 3/4 SMS @ 1" O.C. (On Center).

Dimensions shown:

- 5" MIN (Minimum distance between attachment points)
- 1/2" (Minimum distance from wall to attachment point)

45° - 60°

TOP OF CONCRETE

4" NOMINAL SLAB MINIMUM

3/8" Ø x 3' MASONRY ANCHOR

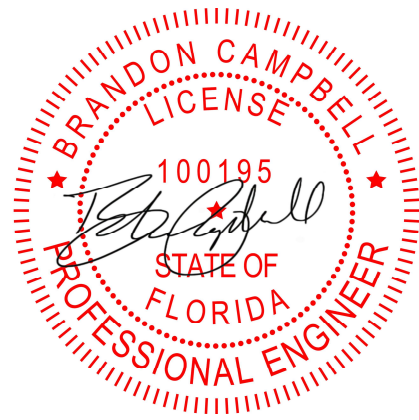
2500 PSI CONCRETE W/WELDED WIRE OR FIBERMESH

1/2" x 3"

2" x 2" x .125' CLIP MAX. OF DECK W/ (2) 1/4" x 1/4" DISTANCE FROM TOP

(2) 1/4" Ø x 3" MASONRY ANCHOR (MIN 2-3/4" EMBEDMENT)

Diagram illustrating the reinforcement details for a column section. The column height is 3'. The reinforcement includes 8" x 8" lineal footing minimum. The lap splice angle is 45° - 60°. The top of concrete is indicated. The reinforcement is labeled as 8" x 3" MIN (NT).



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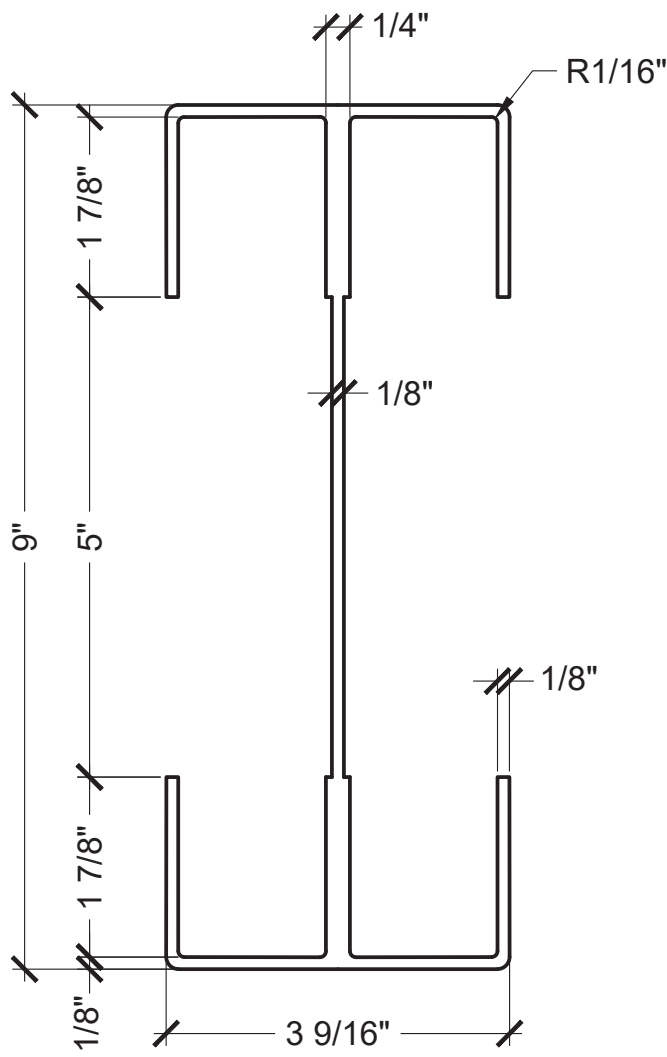
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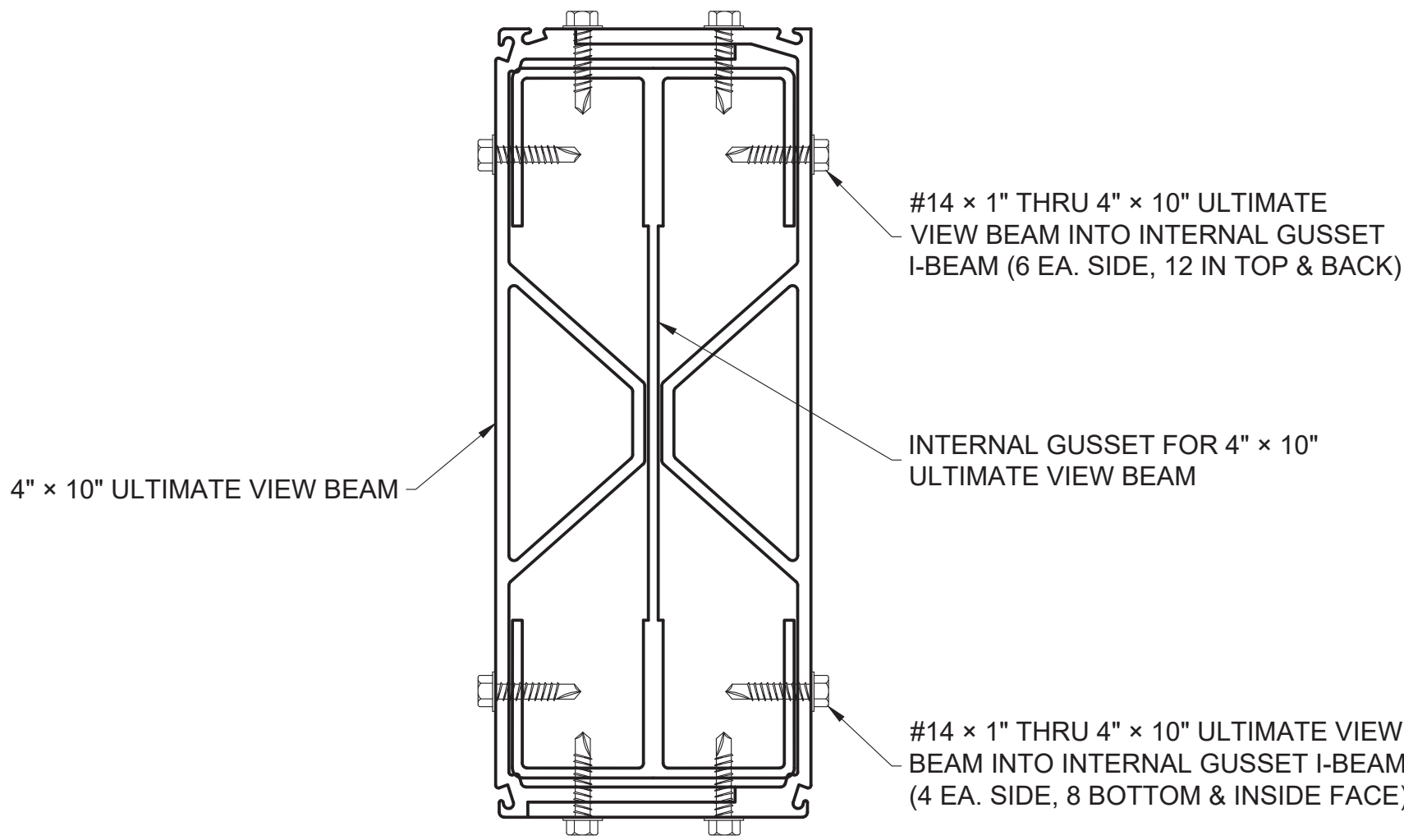
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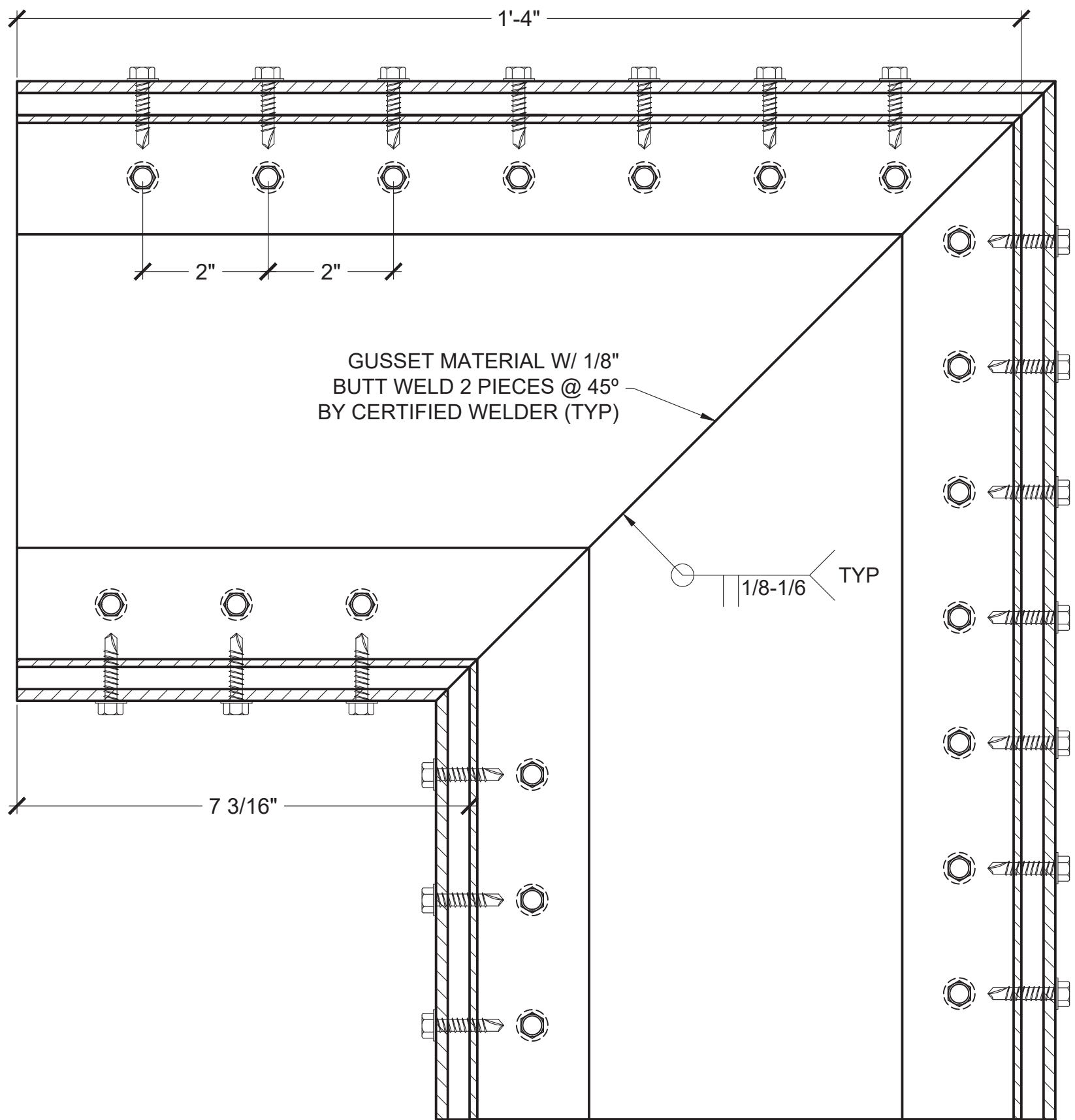




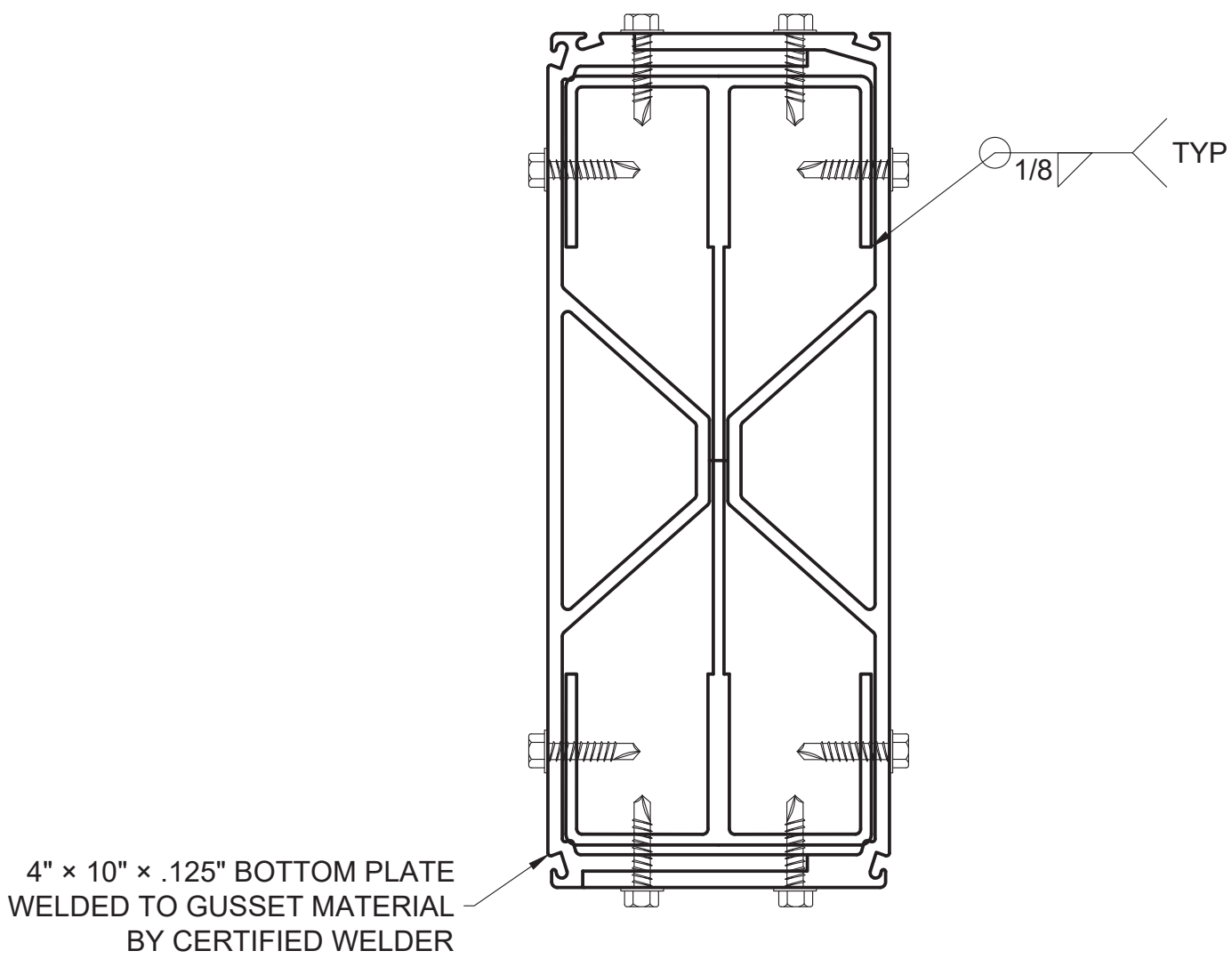
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SCALE: N.T.S.



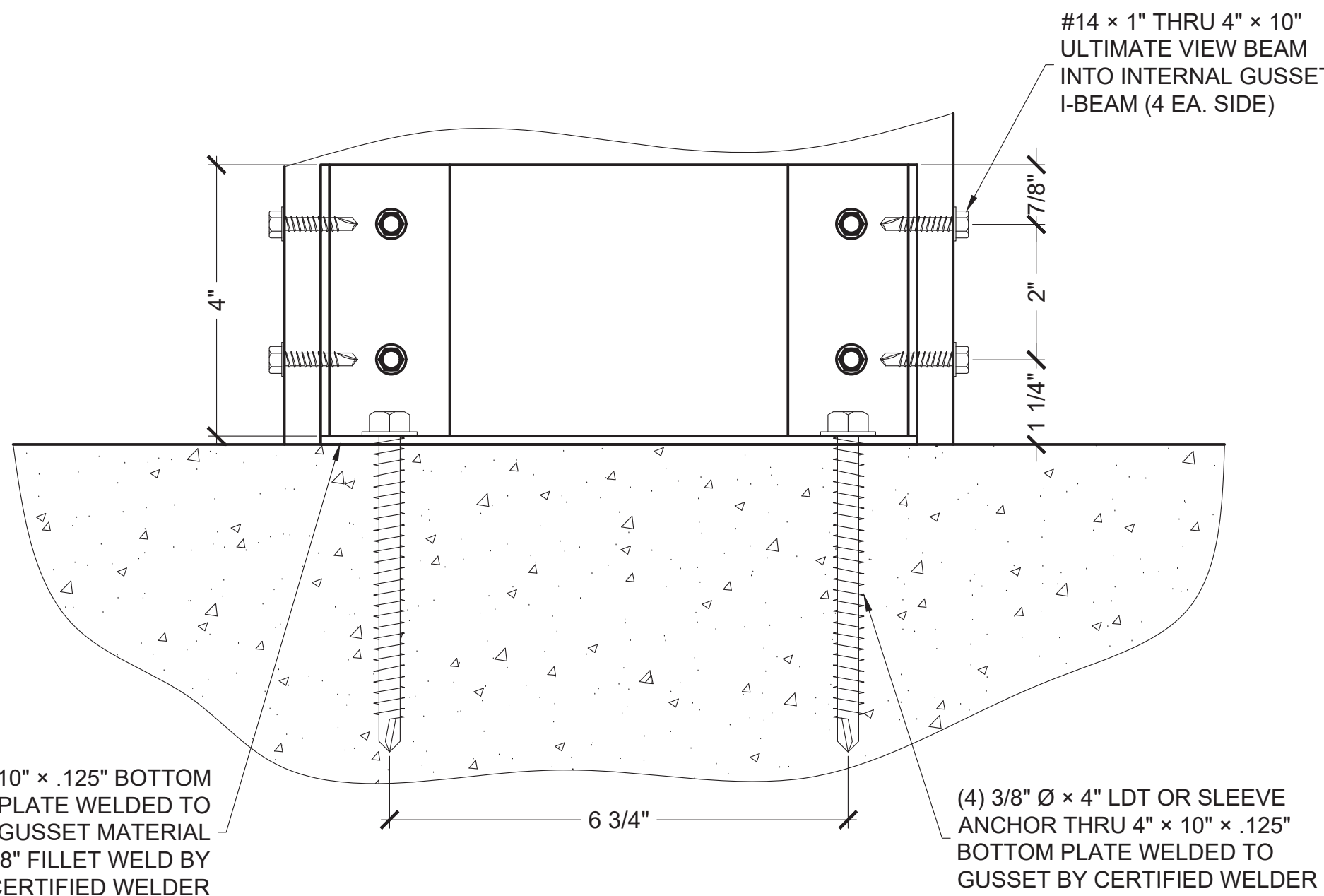
ULTIMATE VIEW UPRIGHT TO BEAM GUSSET MATERIAL  
SCALE: N.T.S.



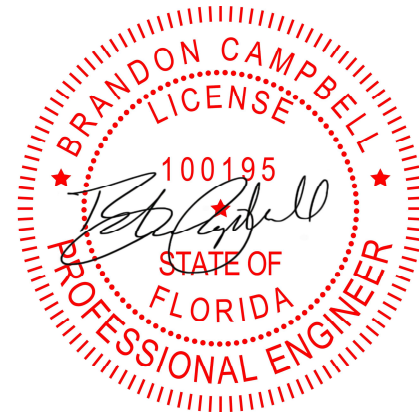
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SCALE: N.T.S.



ULTIMATE VIEW UPRIGHT TO BEAM WELD GUSSET MATERIAL  
SCALE: N.T.S.



ULTIMATE VIEW BASE MATERIAL  
SCALE: N.T.S.



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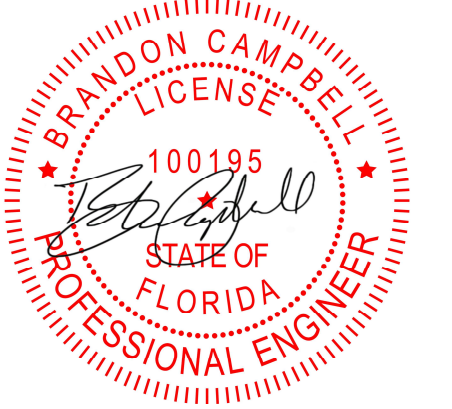
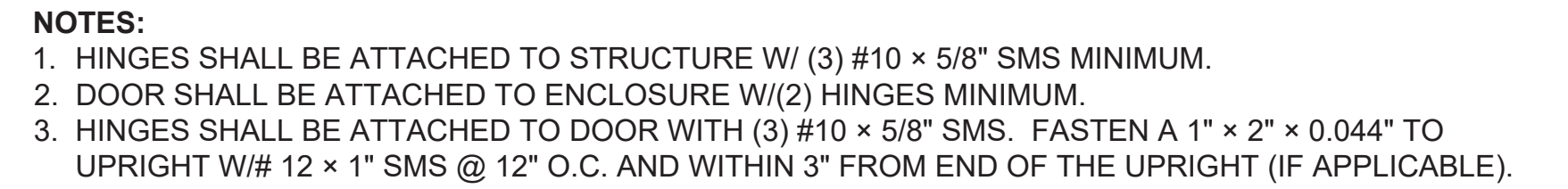
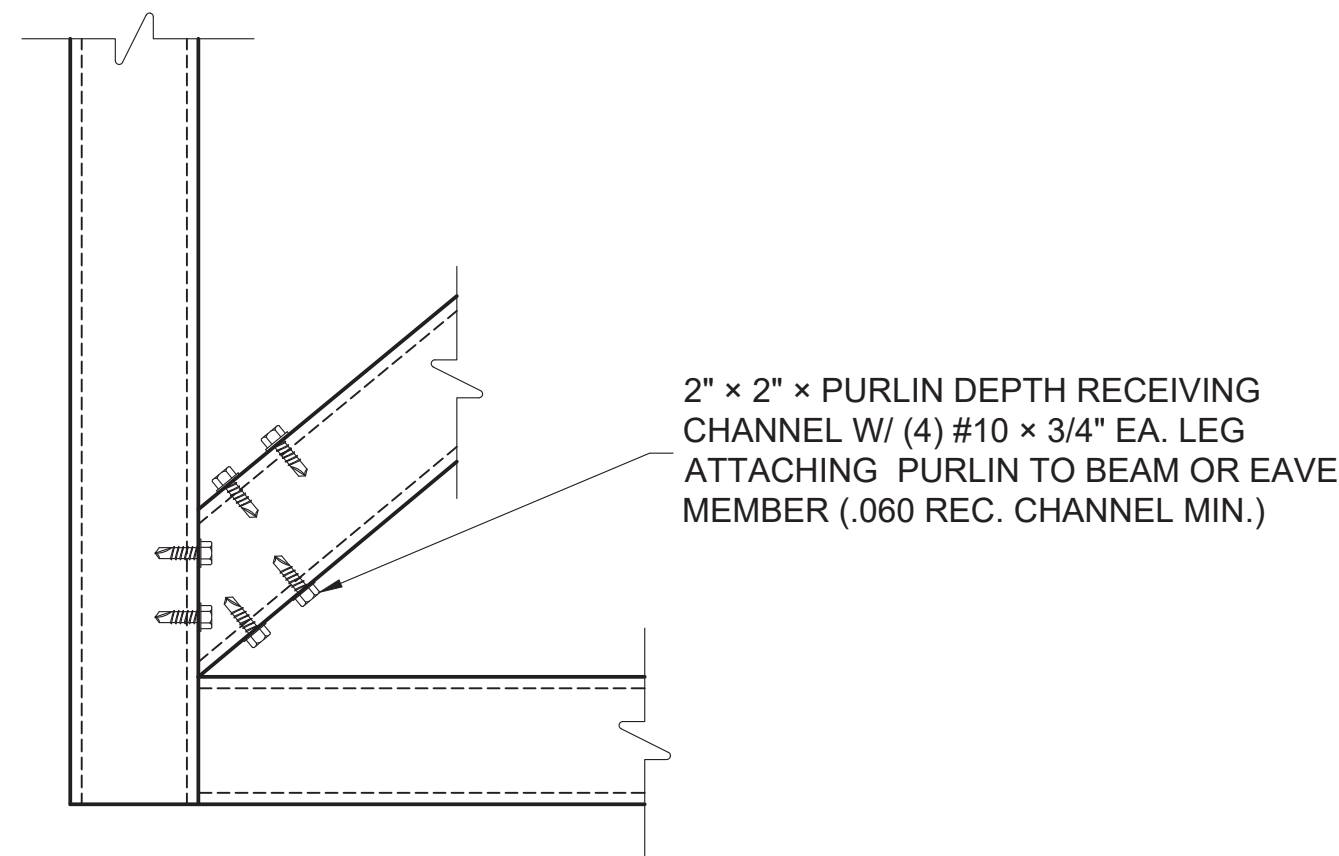
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